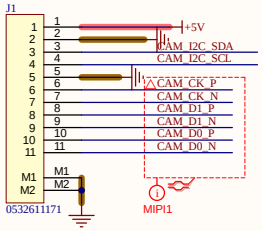
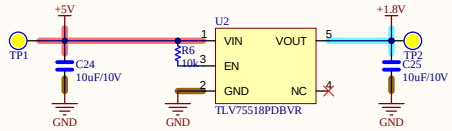
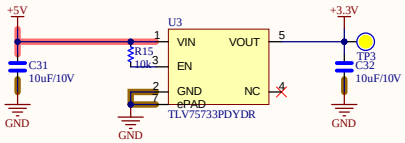
to OBC



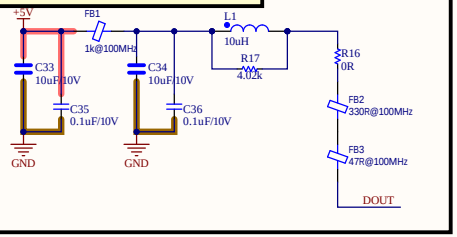
Connector



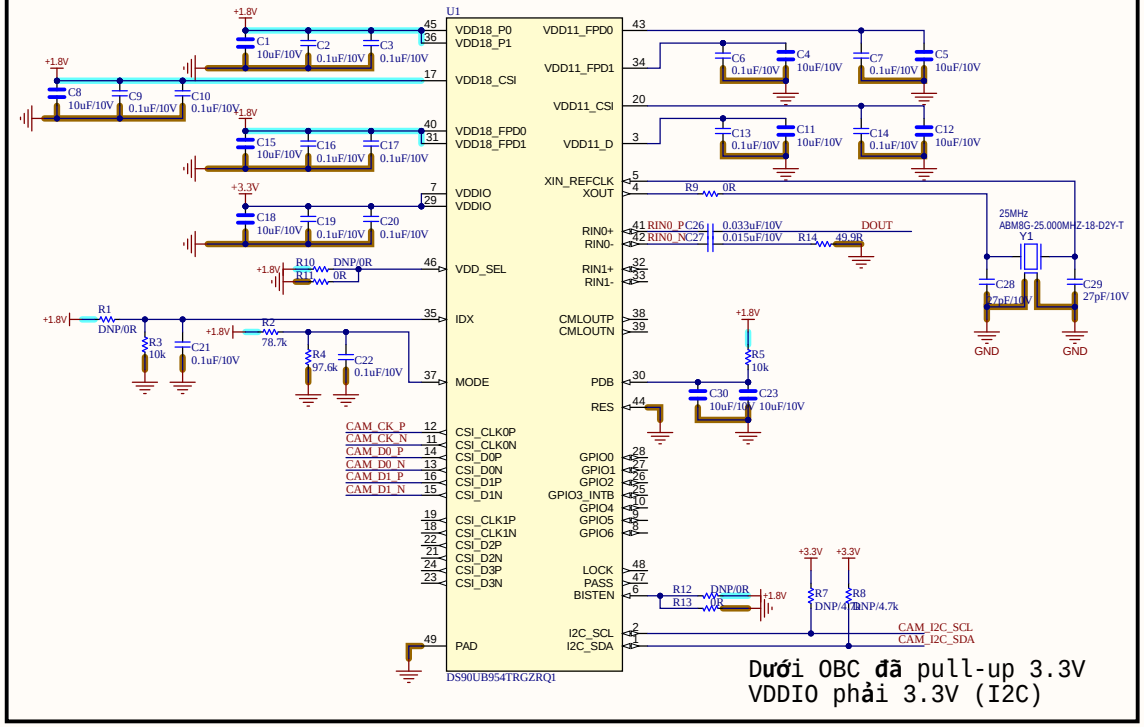
+3.3V POWER SUPPLY



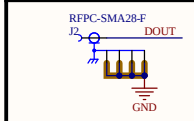
PoC



Deserializer



Connector



Mounting

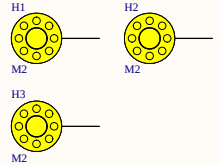


Table 7-2. Strap Configuration Mode Select

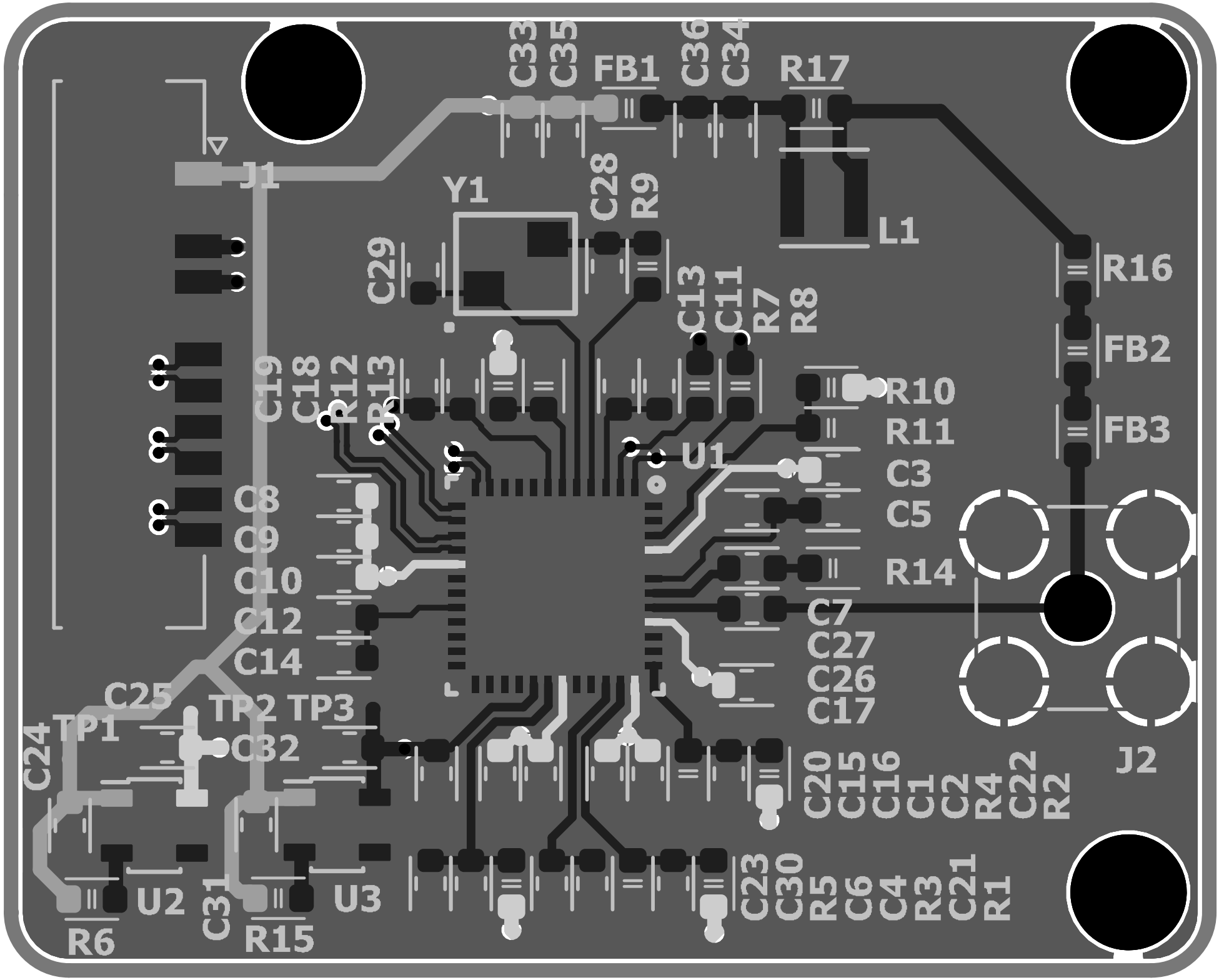
MODE NO.	V_{TARGET} VOLTAGE RANGE			V_{TARGET} STRAP VOLTAGE	SUGGESTED STRAP RESISTORS (1% TOL)		RX MODE
	V_{MIN}	V_{TYP}	V_{MAX}	VDD18 = 1.8 V	R_{HIGH} (kΩ)	R_{LOW} (kΩ)	
0	0	0	0.131 × $V_{(VDD18)}$	0	OPEN	10.0	CSI-2 non-synchronous Back Channel
1	0.179 × $V_{(VDD18)}$	0.213 × $V_{(VDD18)}$	0.247 × $V_{(VDD18)}$	0.374	88.7	23.2	RAW12 LF
	0.642 × $V_{(VDD18)}$	0.673 × $V_{(VDD18)}$	0.704 × $V_{(VDD18)}$	1.202	39.2	78.7	
2	0.296 × $V_{(VDD18)}$	0.330 × $V_{(VDD18)}$	0.362 × $V_{(VDD18)}$	0.582	75.0	35.7	RAW12 HF
	0.761 × $V_{(VDD18)}$	0.792 × $V_{(VDD18)}$	0.823 × $V_{(VDD18)}$	1.420	25.5	95.3	
3	0.412 × $V_{(VDD18)}$	0.443 × $V_{(VDD18)}$	0.474 × $V_{(VDD18)}$	0.792	71.5	56.2	RAW10
	0.876 × $V_{(VDD18)}$	$V_{(VDD18)}$	$V_{(VDD18)}$	1.8	10.0	OPEN	
4	0.525 × $V_{(VDD18)}$	0.559 × $V_{(VDD18)}$	0.592 × $V_{(VDD18)}$	0.995	78.7	97.6	CSI-2 Synchronous Back Channel

Table 7-15. Serial Control Bus Addresses for IDX

NO.	V_{IDX} VOLTAGE RANGE			V_{IDX} TARGET VOLTAGE	SUGGESTED STRAP RESISTORS (1% TOL)		PRIMARY ASSIGNED I2C ADDRESS	
	V_{MIN}	V_{TYP}	V_{MAX}	(V); VDD1P8 = 1.80V	R_{HIGH} (kΩ)	R_{LOW} (kΩ)	7-BIT	8-BIT
0	0	0	0.131 × $V_{(VDD18)}$	0	OPEN	10.0	0x30	0x60
1	0.179 × $V_{(VDD18)}$	0.213 × $V_{(VDD18)}$	0.247 × $V_{(VDD18)}$	0.374	88.7	23.2	0x32	0x64
2	0.296 × $V_{(VDD18)}$	0.330 × $V_{(VDD18)}$	0.362 × $V_{(VDD18)}$	0.582	75.0	35.7	0x34	0x68
3	0.412 × $V_{(VDD18)}$	0.443 × $V_{(VDD18)}$	0.474 × $V_{(VDD18)}$	0.792	71.5	56.2	0x36	0x6C
4	0.525 × $V_{(VDD18)}$	0.559 × $V_{(VDD18)}$	0.592 × $V_{(VDD18)}$	0.995	78.7	97.6	0x38	0x70
5	0.642 × $V_{(VDD18)}$	0.673 × $V_{(VDD18)}$	0.704 × $V_{(VDD18)}$	1.202	39.2	78.7	0x3A	0x74
6	0.761 × $V_{(VDD18)}$	0.792 × $V_{(VDD18)}$	0.823 × $V_{(VDD18)}$	1.420	25.5	95.3	0x3C	0x78
7	0.876 × $V_{(VDD18)}$	$V_{(VDD18)}$	$V_{(VDD18)}$	1.8	10.0	OPEN	0x3D	0x7A

Space ChinTech

Project:	Sub-system:	
Design by: *	Sheet title: Deserializer.SchDoc	Sheet 1 of 1
Checked by: *	Document No: 1	Size: A3
Approved by: *	Revision: *	Date: *


C33
C35
FB1
C36
C34
R17
J1
Y1
C28
R9
L1
C29
C13
C11
R7
R8
R16
FB2
FB3
C19
C18
R12
R13
R10
R11
U1
C3
C5
C8
C9
C10
C12
C14
R14
C7
C27
C26
C17
C25
TP2
TP3
TP1
C32
C24
J2
C20
C15
C16
C1
C2
R4
C22
R2
U2
U3
C31
R6
R15
C23
C30
R5
C6
C4
R3
C21
R1

Name	Description	Designator	Footprint	LibRef	Quantity
10uF/10V	CAP High Value	C1, C4, C5, C8, C11, C12, C15, C18, C23, C24, C25, C30, C31, C32, C33, C34	CAP_C_1608X80	_CAP_C_High_2	16
0.1uF/10V	CAP	C2, C3, C6, C7, C9, C10, C13, C14, C16, C17, C19, C20, C21, C22, C35, C36	CAP_C_1608X80	_CAP_C_2	16
0.033uF/10V	CAP	C26	CAP_C_1608X80	_CAP_C_2	1
0.015uF/10V	CAP	C27	CAP_C_1608X80	_CAP_C_2	1
27pF/10V	CAP	C28, C29	CAP_C_1608X80	_CAP_C_2	2
1k@100MHz	FERRITE BEAD 600 OHM 0603 1LN	FB1	IND_C_1608X60	BLM18AG601SN1D	1
330R@100MHz	FERRITE BEAD 600 OHM 0603 1LN	FB2	IND_C_1608X60	BLM18AG601SN1D	1
47R@100MHz	FERRITE BEAD 600 OHM 0603 1LN	FB3	IND_C_1608X60	BLM18AG601SN1D	1
M2		H1, H2, H3	HEX-2.2mm	HEX-M3	3
0532611171		J1	MOLEX_0532611171	Header 11x1_2M	1
RFPC-SMA28-F	SMA Connector Jack, Female Socket 50 Ohms Panel Mount, Through Hole Solder	J2	GRADCONN_RFPC-SMA28-F	RFPC-SMA28-F	1
10uH		L1	IND300X300X150	_IND	1
DNP/0R	RES	R1, R10, R12	RES_C_1608X60	_RES_B_2	3
78.7k	RES	R2	RES_C_1608X60	_RES_B_2	1
10k	RES	R3, R5, R6, R15	RES_C_1608X60	_RES_B_2	4
97.6k	RES	R4	RES_C_1608X60	_RES_B_2	1
DNP/4.7k	RES	R7, R8	RES_C_1608X60	_RES_B_2	2
0R	RES	R9, R11, R13, R16	RES_C_1608X60	_RES_B_2	4
49.9R	RES	R14	RES_C_1608X60	_RES_B_2	1
4.02k	RES	R17	RES_C_1608X60	_RES_B_2	1
DS90UB954TRGZRQ1	Dual 2 MP FPD-Link III deserializer with CSI-2 outputs for 2MP/60fps cameras & radar 48-VQFN - 40 to 105	U1	QFN50P700X700X100-48N-EP410x410	IC_Deserializer_DS90UB954TRGZRQ1	1
TLV75518PDBVR		U2	SOT25	IC_POWER_AP7347DQ	1
TLV75733PDYDR		U3	SOT25-EPAD	IC_POWER_TLV75733PDYDR	1
Crystal 4 Pins	Crystal	Y1	CRYSTAL_ABM8G	OSC_24MHz	1